

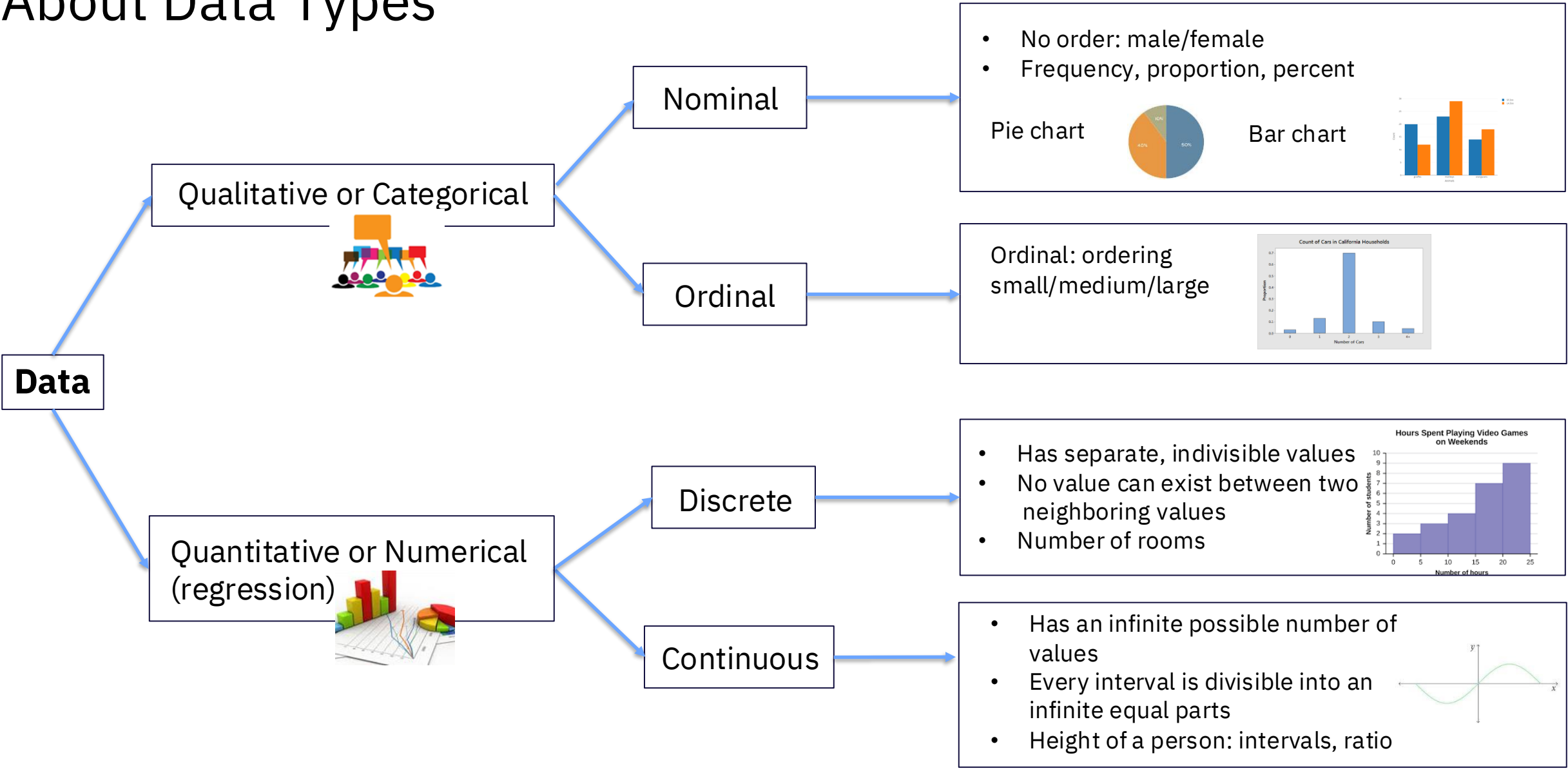
About Data



1. Data Types
2. What is so confusing about the confusion matrix
3. Resources



About Data Types



Characteristics of data

The 5 Vs

Volume

Refers to the vast amounts of data generated every second.

Variety

Refers to the different types of data we can now use.

Velocity

Refers to the speed at which new data is generated and the speed at which data moves around.

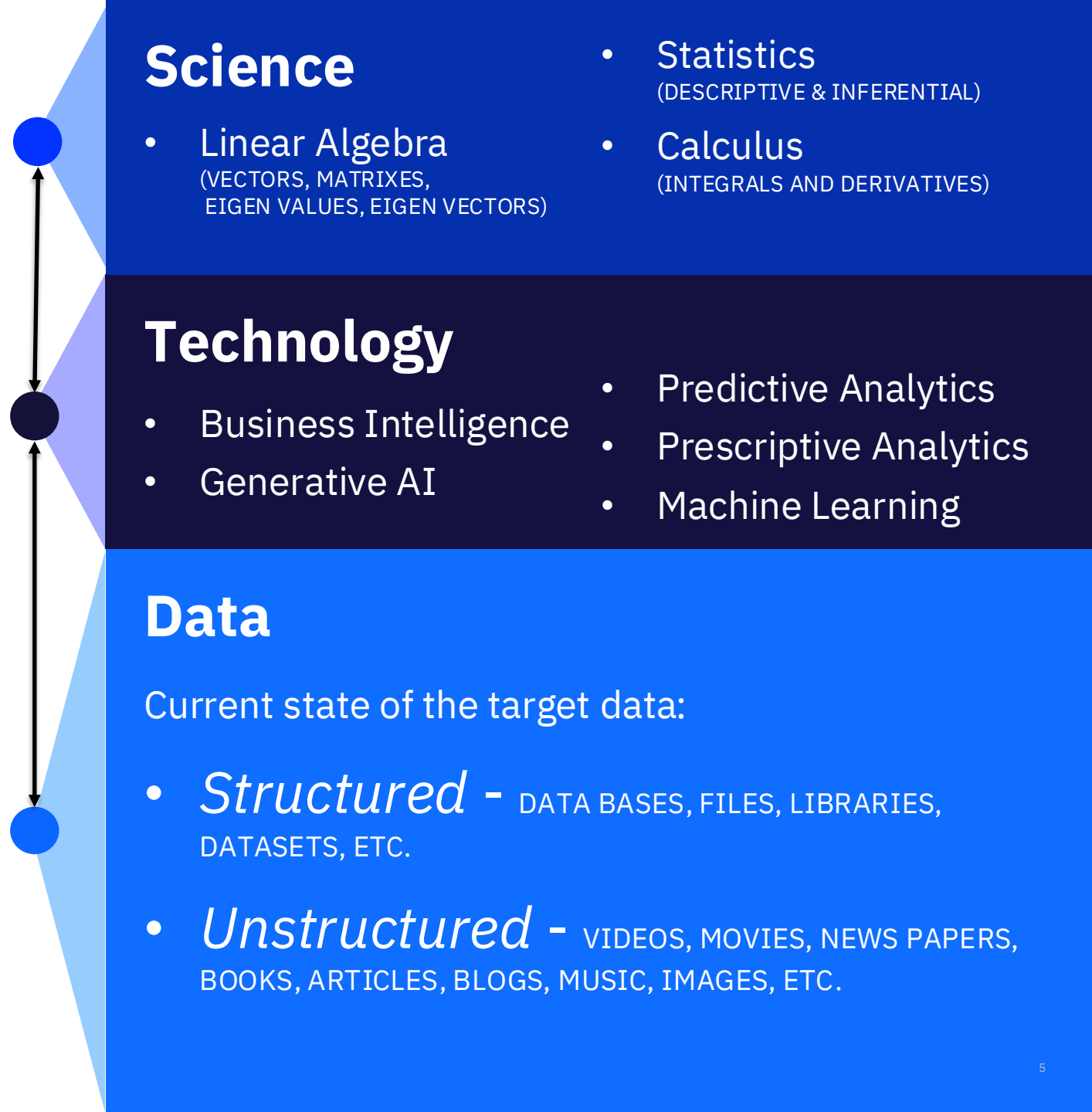
Veracity

Refers to the messiness or trustworthiness of the data.

Value

Refers to having access to big data is no good unless we can turn it into value.

Science,
Technology
and **Data** are
intrinsically
connected



Data cleansing

Data **Analysts** can spend up to **80% of the time cleaning data.**

- Importing data
- Joining multiple datasets
- Detecting missing values
- Detecting anomalies
- Imputing for missing values
- Data quality assurance

Tidy data lends itself to efficient data analysis and processing

“Tidy datasets are all alike, but every messy dataset is messy in its own way.”

– Hadley Wickham

- **Transforming your data into standard format**, or tidy data, makes analysis and storage easier down the road
- Additionally, one must make sure the data are in its **appropriate type**

What is tidy data?

Tidy data satisfies 3 components

1. Each variable forms a column.

Brand	Size	Price
McDonald's	Small	1.00
Dunkin Donuts	Small	1.99
Starbucks	Small	2.99
McDonald's	Medium	1.00
Dunkin Donuts	Small	2.49
Starbucks	Medium	3.49

2. Each observation forms a row.

Brand	Size	Price
McDonald's	Small	1.00
Dunkin Donuts	Small	1.99
Starbucks	Small	2.99
McDonald's	Medium	1.00
Dunkin Donuts	Small	2.49
Starbucks	Medium	3.49

3. Each type of observational unit forms a table.

Brand	Size	Price
McDonald's	Small	1.00
Dunkin Donuts	Small	1.99
Starbucks	Small	2.99
McDonald's	Medium	1.00
Dunkin Donuts	Small	2.49
Starbucks	Medium	3.49

Example of tidy data

Not so tidy.....

Brand	Size	Price
McDonald's	Small	1.00
Dunkin Donuts	Small	1.99
Starbucks	Small	2.99
McDonald's	Medium	1.00
Dunkin Donuts	Medium	2.49
Starbucks	Medium	3.49

Tidy data....also known as feature engineering

Size	McDonald's	Dunkin Donuts	Starbucks
Small	1.00	1.99	2.99
Medium	1.00	2.49	3.99

From outlier checking to feature engineering to missing value imputation

Column headers are values, not variable names.

Multiple variables are stored in one column.

Variables are stored in both rows and columns.

Multiple types of observational units are stored in the same table.

A single observational unit is stored in multiple tables.

Missing Data

Explicit

- Marked with NULL or NA
- Use summary functions, eg, `summary()`, as its output will list each variable's count of missing values

Implicit

- Not there at all
- You must explore and visualize your data to notice things that appear 'off'

Methods for handling missing data

Remove the observation completely

Imputation methods

- Replace with summary statistic such as mean, median, or mode
- Create a new variable that flags a missing column
- Replace NA with an outlier (tree-based models will implicitly understand that these outliers are associated with missing values)

Statistical analysis

The data representation phase should leverage mathematical tools such as:

- **Statistics**
- **Correlations**
- **Chi-squared tests**

Descriptive statistics allow you to describe a vast, complex data set using just a few key numbers.

You can also create a table that displays summarized statistics for cases grouped by categorical data based on a single measure.

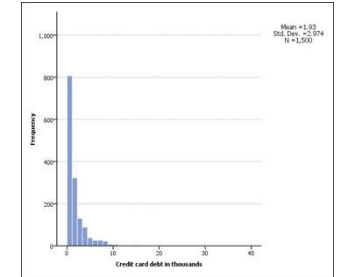
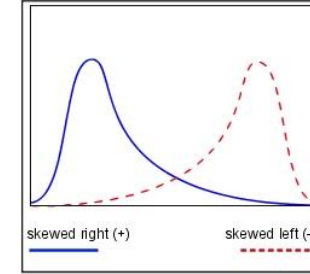
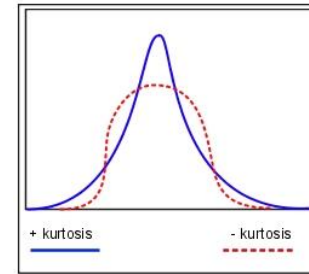
The table below shows the mean household income for customers grouped by education level.

Descriptive Statistics					
	High school degree	Post-undergraduate degree	Did not complete high school	Some college	College degree
Mean	52.00	99.71	51.48	56.90	70.94
Std. Deviation	56.370	147.769	51.855	53.836	67.940
N	527	84	246	333	310
Median	35.00	59.50	36.00	39.00	49.00
Minimum	12	16	15	13	15
Maximum	533	1,079	497	403	512

Descriptive Statistics Quantitatively Summarize a Data Set

You can use descriptive statistics to:

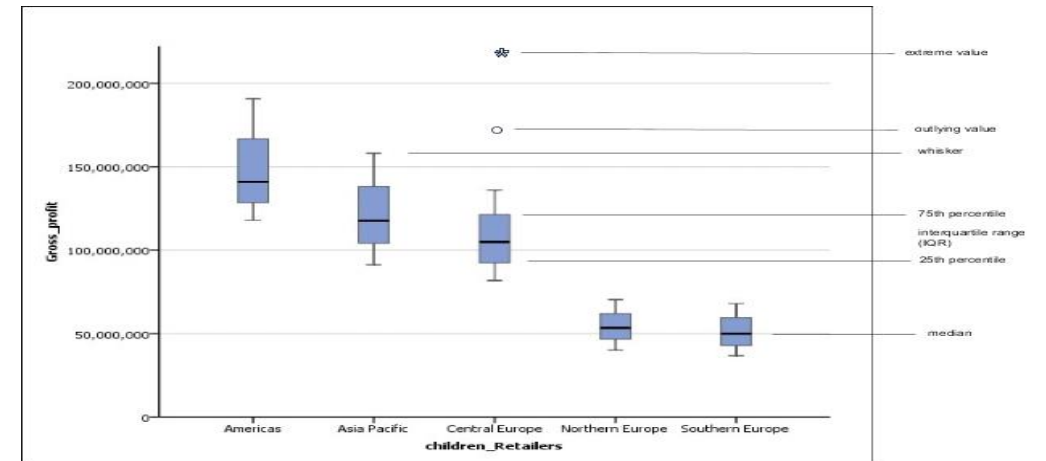
- Look at averages, such as the **mean** or **median**.
- Obtain information, such as the **mean for groups of interest**, that you might need to interpret other statistical tests.
- Provide graphical representations of data, such as **histograms** and **boxplots**.



Descriptive Tables

- Measures of Central Tendency
- Measures of Dispersion
- Measures of Distribution

Histograms



Boxplots

Variance vs Standard Deviation

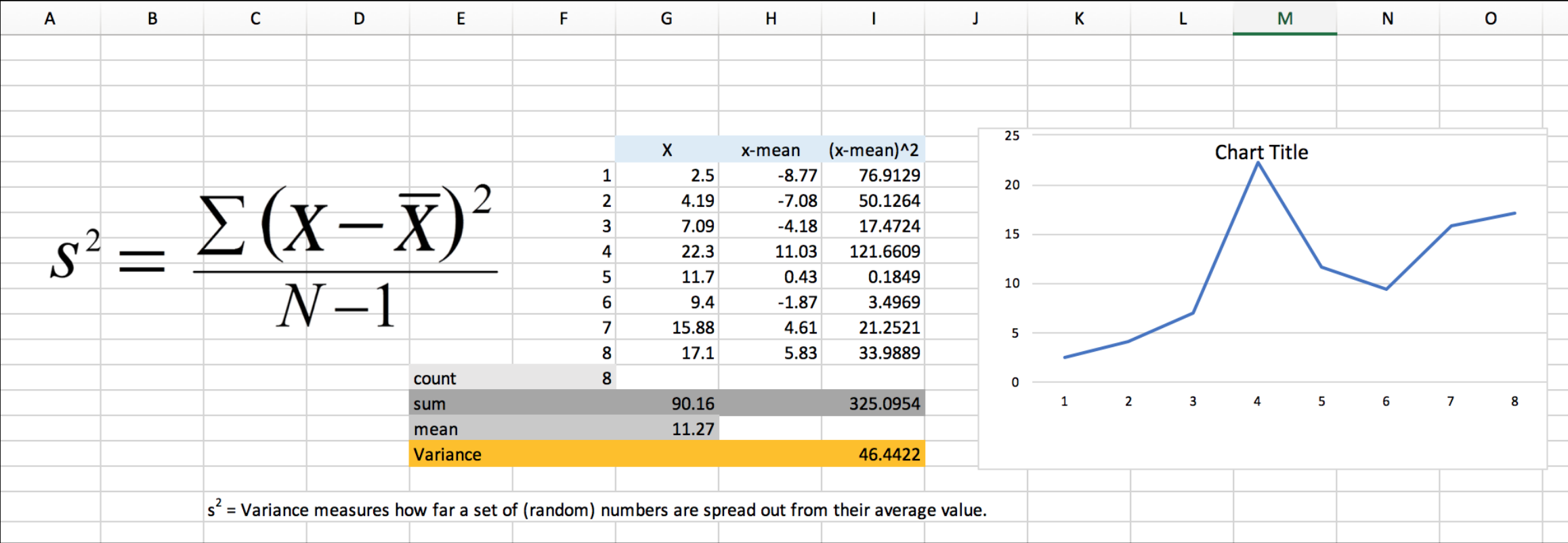
Variance measures the average degree to which each point differs from the mean.

The greater the variance, the larger the overall data range. It is a good way to spot the outliers and gives you an idea of the **overall spread**.

Standard deviation is the square root of the variance.

The calculation of variance uses squares because it weights outliers more heavily than data very near the mean.

It starts with calculating the variance



Risk of using summary statistics

There's a **danger** in relying only on summary statistics and **ignoring the overall distribution.**

Anscombe's quartet offers a classic example of this risk.

It comprises four data sets of 11 x, y points that have nearly identical simple statistical properties, yet appear very different when graphed.

Anscombe's Quartet

B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	
Sources:	Edward R. Tufte, <i>The Visual Display of Quantitative Information</i> (Cheshire, Connecticut: Graphics Press, 1983), pp. 14-15. F.J. Anscombe, "Graphs in Statistical Analysis," <i>American Statistician</i> , vol. 27 (Feb 1973), pp. 17-21.																	
	Observation	x1	y1			x2	y2			x3	y3			x4	y4			
	1	10	8.04			10	9.14			10	7.46			8	6.58			
	2	8	6.95			8	8.14			8	6.77			8	5.76			
	3	13	7.58			13	8.74			13	12.74			8	7.71			
	4	9	8.81			9	8.77			9	7.11			8	8.84			
	5	11	8.33			11	9.26			11	7.81			8	8.47			
	6	14	9.96			14	8.1			14	8.84			8	7.04			
	7	6	7.24			6	6.13			6	6.08			8	5.25			
	8	4	4.26			4	3.1			4	5.39			19	12.5			
	9	12	10.84			12	9.13			12	8.15			8	5.56			
	10	7	4.82			7	7.26			7	6.42			8	7.91			
	11	5	5.68			5	4.74			5	5.73			8	6.89			
Sum			99	82.51			99	82.51			99	82.5			99	82.51		
count(n)		11																
average (mean)			9	7.50090909			9	7.50090909			9	7.5			9	7.50090909		
variance s ²			11	4.12726909			11	4.1272909			11	4.12262			11	4.12324909		

Using representation to help us detect bad data

Univariate analysis:

- Histograms allow one to understand the distribution of a variable

Bivariate analysis:

- Bar charts and Box Plots can help an analyst compare groups
- Line charts help an analyst understand trends over time
- Scatterplots are best used to understand relationships of values

Categorical variables must be mapped to a number in order to be used by a machine learning model

Nominal variables:

- colors
- animal species
- countries

Ordinal:

- rankings
- socioeconomic status

How do you represent categories as numbers?

Naïve approach:

Map a category to a number: [red, blue, green] = [1, 2, 3]

This is misleading!

By mapping the data like this, you are implying green is 3x greater than red.

1. Data Types
2. **What is so confusing about the Confusion Matrix**
3. Resources



How important is Accuracy?

The client asks:

“How do I get the most accurate model?”

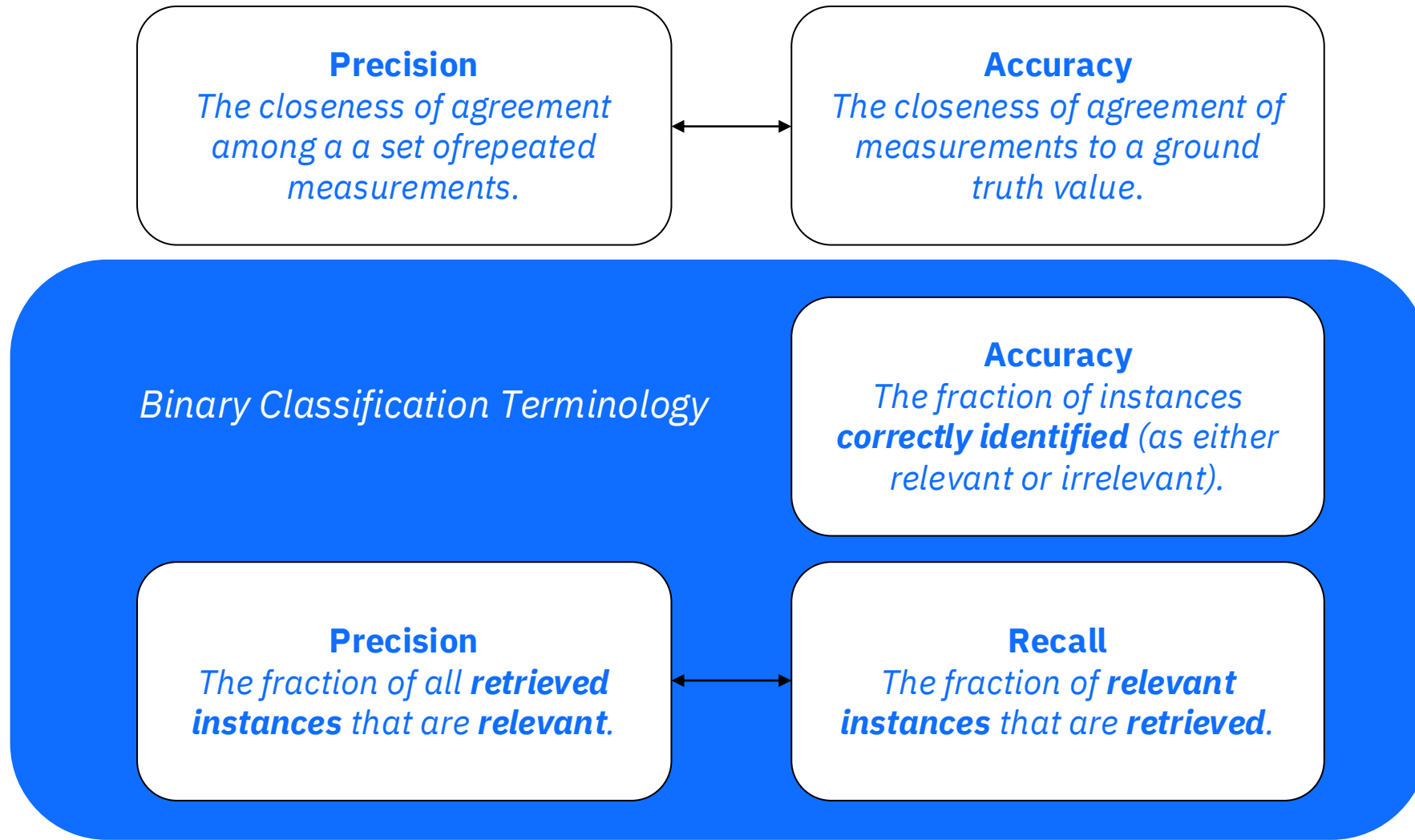
The data scientist responds by asking:

“What business challenge are you trying to solve using the model?”

So Accuracy need not necessarily be the one-and-only model metrics data scientists chase and include simple explanation of other metrics as well.

What about precision and recall?

What About Precision and Recall?



When recall is more important than precision

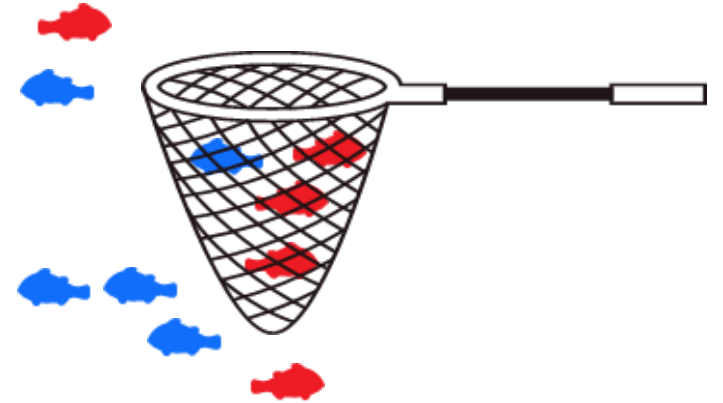
5 Relevant data points (red fish)

5 Irrelevant data points (blue fish)

The search has retrieved 3 relevant documents out of a total of 5 relevant data points from the data set and 1 irrelevant document.

Recall = $3 / 5 = 0.6$

Precision = $3 / 4 = 0.75$ (the blue fish is not part of the equation at all).



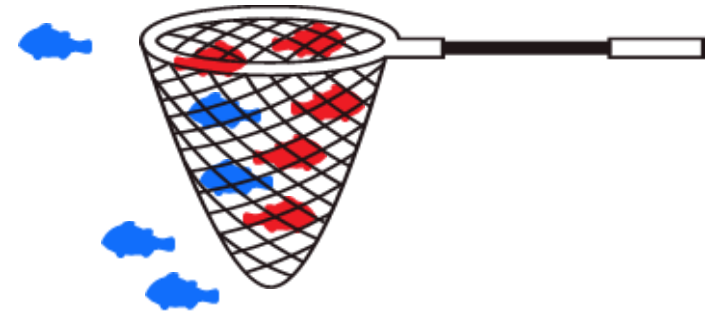
The case of 100% recall and low precision

5 Relevant data points (red fish)

5 Irrelevant data points (blue fish)

In some models, this is the preferred scenario even though there may be some irrelevant data points with a high score.

The algorithm team will then work on increasing the precision of this system.



The case of 100% precision and low recall

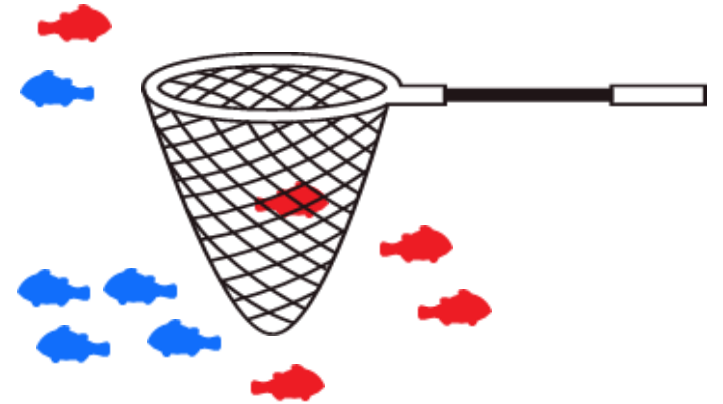
5 Relevant data points (red fish)

5 Irrelevant data points (blue fish)

Zero false positives, 100% precision — no blue fish in the net

But there are many false negatives — many red fish in the sea

There are potentially many data points that we will never consider. Perfect precision with poor recall is of no value to any ML model.



False Positives can be Dangerous

Notice that the accuracy for this model is very high, at 99.9%!!

But what if I mentioned that the positive over here is actually someone who is sick and carrying a virus that can spread very quickly?

Or the positive here represent a fraud case?

Or the positive here represents terrorist that the model says its a non-terrorist?

		Predicted/Classified	
		Negative	Positive
Actual	Negative	998	0
	Positive	1	1

Precision and Recall

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

		Predicted	
		Negative	Positive
Actual	Negative	True Negative	False Positive
	Positive	False Negative	True Positive

Precision

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

		Predicted	
		Negative	Positive
Actual	Negative	True Negative	False Positive
	Positive	False Negative	True Positive

Recall

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$
$$= \frac{\text{True Positive}}{\text{Total Actual Positive}}$$

		Predicted	
		Negative	Positive
Actual	Negative	True Negative	False Positive
	Positive	False Negative	True Positive

May the Best Model Win

Recall actually calculates how many of the Actual Positives our model captures through labeling it as Positive (True Positive).

Recall shall be the model metric we use to select our best model when there is a high cost associated with False Negative.

In fraud detection or sick patient detection:

If a fraudulent transaction (Actual Positive) is predicted as non-fraudulent (Predicted Negative), the consequence can be very bad for the bank.

Deriving Precision, Recall, and Accuracy from the Confusion Matrix

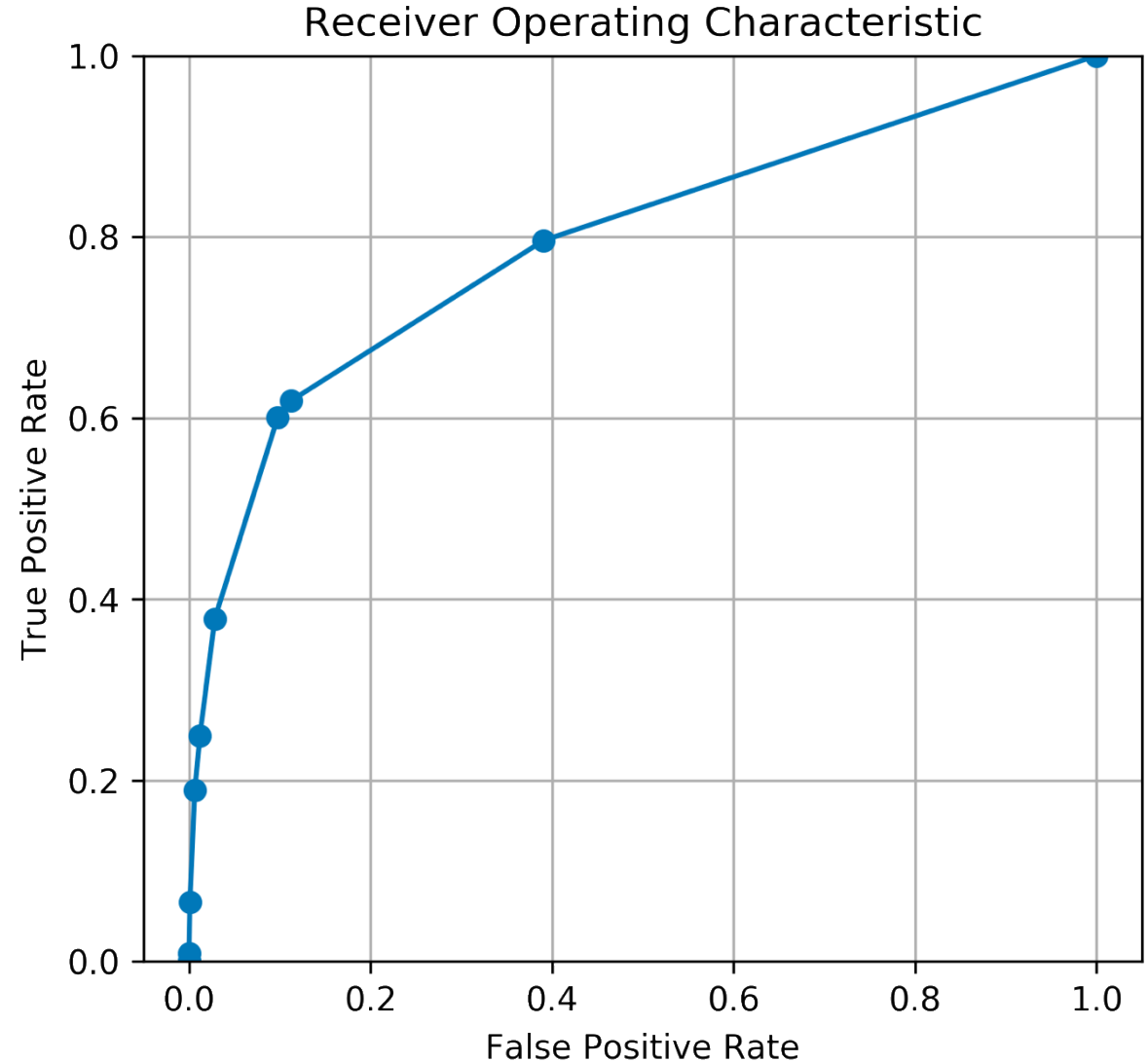
Accuracy = -----

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy

May the Best Model Win

The ***True Positive Rate*** is sometimes plotted against the ***False Positive Rate*** in order to see the tradeoff between the two as thresholds are adjusted.

This plot is called the ***Receiver Operating Characteristic***.

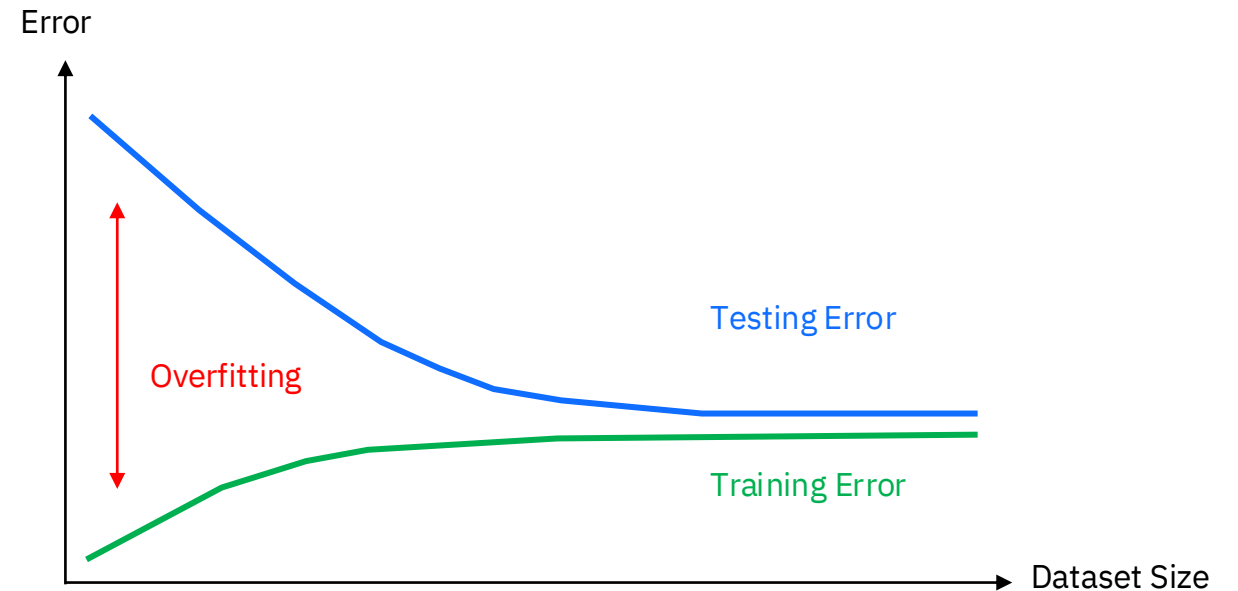


Dangers of Overfitting Data

The biggest risk is overfitting

- A model that works well on training data but performs badly on new data

Trick is to select a model without knowing the data it will be applied to



Example of overfitting

We want to predict
if a student with a
GPA of 3.3 will be
admitted to
Wentworth Institute
of Technology in
Boston

Assume we train a model from a dataset of 5,000 students and their outcomes.

Next, we try the model out on the original dataset, and it predicts outcomes with 99% accuracy... wow!

But now comes the bad news.

When we run the model on a new (“unseen”) dataset of student admissions, we only get 50% accuracy... uh-oh!

Our model doesn’t *generalize* well from our training data to unseen data.

This is known as overfitting, and it’s a common problem in machine learning and data science.

F1 Score

F1 is a function of
Precision and Recall

$$F1 = 2 \times \frac{Precision * Recall}{Precision + Recall}$$

**F1 Score is needed when you want to seek
a balance between Precision and Recall.**

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- ▶ 3. Resources



Resources

- [1] Beyond the hype: A guide to understanding and successfully implementing artificial intelligence within your business
<https://www.ibm.com/downloads/cas/8ZDXNKQ4>
- [2] A Practical Guide to Building Enterprise Applications: by Tom Markiewicz and Josh Zheng – Feb 2018 O'Reilly
<https://tmarkiewicz.com/getting-started-with-artificial-intelligence/>
- [3] The New York Times - Nils Nilsson
<https://www.nytimes.com/2019/04/25/obituaries/nils-nilssen-dead.html>
- [4] Why artificial intelligence is enjoying a renaissance
<http://www.economist.com/blogs/economist-explains/2016/07/economist-explains-11>
- [5] How Cognitive Systems Could Redefine The Way Governments Work
<http://www.forbes.com/sites/ibm/2016/09/20/how-cognitive-systems-could-redefine-the-way-governments-work/#1e1ed4f52ff1>
- [6] March of the Machines
<http://www.economist.com/news/leaders/21701119-what-history-tells-us-about-future-artificial-intelligenceand-how-society-should>
- [7] We have to upgrade our skills to match intelligent machines
<http://www.businessinsider.com/how-labor-can-keep-up-with-artificial-intelligence-2016-10?IR=T>
- [8] Forbes & IBM - Intelligent Automation: How AI and Automation are Changing the Way Work Gets Done
<https://www.ibm.com/downloads/cas/RE2XMOLR>